# FINOPS-01: DPS Capability Units and Querying Consumption with DQL

> **Series:** FINOPS — Cost Management & FinOps | **Reference:** 01 — DPS Capability Units and Querying Consumption with DQL | **Created:** May 2026 | **Last Updated:** 10/14/2026

## Overview

*"How much are we using, and where is it going?"* — the single most common DPS question, and the one customers most often answer by squinting at the Subscription portal once a month. This entry is the platform-engineer view of the same question: how DPS consumption is recorded inside the tenant, where the data lives in Grail, what the DQL canon looks like, and how to reconcile in-tenant numbers against the Account Management portal.

**Two surfaces, one truth.** Consumption data appears in two places: as raw per-record `BILLING_USAGE_EVENT` records in `dt.system.events`, and as pre-aggregated `dt.billing.*` metric series. They report the same underlying usage, but at different granularity — choose based on whether you need per-record attribution or fast time-aligned aggregates.

**Per-capability schemas.** There is no universal `billed_*` field. Each capability has its own unit field — `billed_gibibyte_hours` for Full-Stack, `billed_bytes` for log ingest, `data_points` for metrics, `billed_sessions` for RUM, and so on. Mixing units across capabilities is the most common authoring mistake. § 2 lists the unit for each capability.

> **Scope:** Dynatrace SaaS, DPS license model. Classic licensing surfaces some of the same consumption data — the linked Classic-license doc covers the equivalents. Anything called out as **Softened** evolves sprint-to-sprint; verify against current docs before signing off on a procurement review.

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## Table of Contents


1. Short Answer
2. The DPS Capability Model
3. Two Data Surfaces — `dt.system.events` and `dt.billing.*`
4. Mandatory Patterns — `dedup`, `event.kind`, `billing_type`
5. Querying Host-Based Capabilities
6. Querying Byte-Based Capabilities
7. Querying Count-Based Capabilities
8. Metrics-Ingest with Included-Quota Subtraction
9. Query-Side Billing — Chargeback by App and User
10. Per-Bucket and Per-Cost-Center Attribution
11. Common Pitfalls
12. Validating Numbers Against Account Management
13. Bootstrap from the Ready-Made Usage Dashboards
14. Recommended Approach
15. Summary and Next Steps


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## Prerequisites

| Requirement | Details |
|---|---|
| **Dynatrace Environment** | SaaS on the Dynatrace Platform Subscription (DPS). Most queries also work on Classic licensing — output reflects DPS-equivalent usage. |
| **Permissions** | `storage:events:read` for `fetch dt.system.events`; `storage:metrics:read` for the `dt.billing.*` metric series; `account-management` role at the account level for cross-checking against the Subscription portal. |
| **Audience** | Platform / Observability Lead (primary); Executive / Procurement (TL;DR and §12). |

| Requirement | Details |
| --- | --- |
| **Related series** | ORGNZ (bucket strategy, retention), OPLOGS (sampling and ingestion control), AUTOM (Cost Allocation app deployment), ADOPT-05 (optimization roadmap and ROI framing). |
| **Companion FINOPS entries** | FINOPS-02 (forecasting + anomaly detection on consumption), FINOPS-03 (the Cut / Tune / Filter optimization decision framework). |

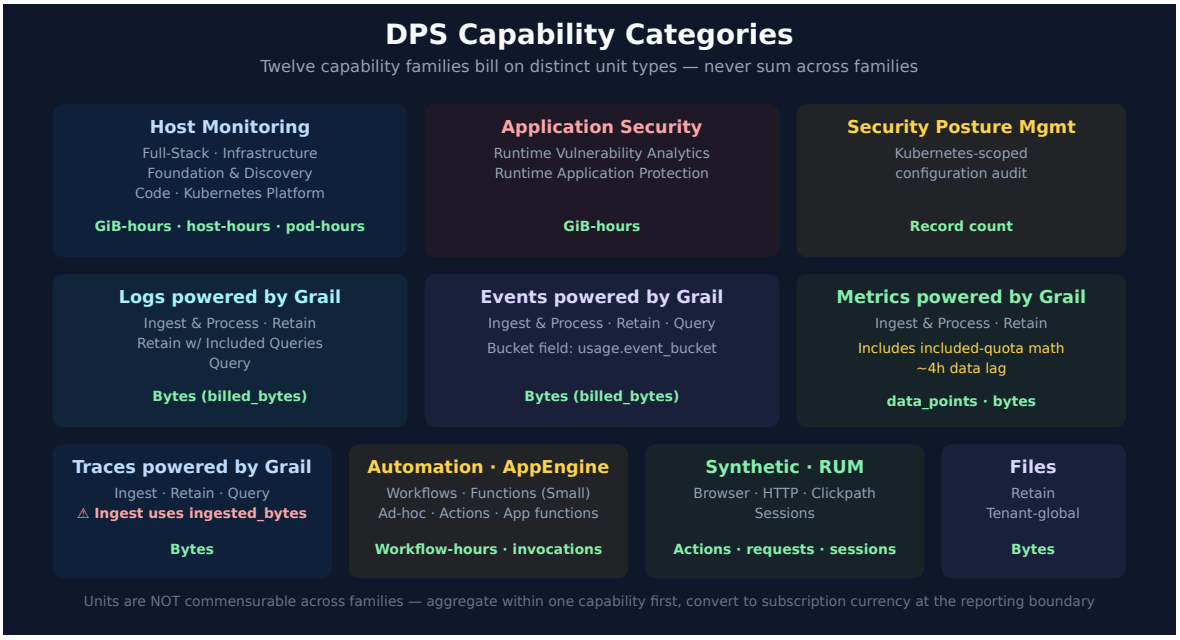
# 1. Short Answer

| Question | One-line answer |
| --- | --- |
| Where does consumption data live? | Per-record: `fetch dt.system.events \| filter event.kind == "BILLING_USAGE_EVENT"`. Pre-aggregated: `timeseries <metric> (dt.billing.<capability>.usage)`. |
| What's the unit field? | Per capability — 7 distinct unit-field families. `billed_gibibyte_hours`, `billed_container_hours`, `billed_pod_hours`, `billed_bytes`, `ingested_bytes` (Traces-Ingest only), `data_points` (Metrics), `billed_synthetic_action_count`, `billed_http_request_count`, `billed_sessions`, `billed_invocations`, or record-count (Workflows). |
| What's the canonical filter pattern? | `event.kind == "BILLING_USAGE_EVENT"` then `dedup event.id` before any aggregation. |
| How do I attribute cost to a team or product? | Use the pre-aggregated `dt.billing.*_by_costcenter` / `_by_product` metric series (Logs and Traces only), or `expand dt.cost.costcenter` / `expand dt.cost.product` on the per-record events. Other capabilities attribute via `usage.bucket`, `dt.entity.host` (moving to `dt.smartscape.host` with SaaS 1.347), or `dt.entity.application`. |
| How fresh is the data? | Most capabilities update every 15 minutes (`usage.start` / `usage.end`). Metrics - Ingest lags by ~4 hours, so query timeframes shorter than 4 hours are incomplete. Automation Workflow uses 1-hour windows. |
| What's the account-level equivalent? | Account Management portal → Subscription → Cost Overview (billable totals), Budget Alerts (commit tracking), Cost Monitors (anomaly detection). The portal applies subscription-currency conversion + reconciliation logic that DQL does not. |

| Question | One-line answer |
|---|---|
| When does DQL disagree with the portal? | Three typical causes: (a) portal converts raw units to subscription currency, DQL does not; (b) portal includes pending reconciliations; (c) time-zone or time-window alignment. See §12. |

> **Sources:** Dynatrace Platform Subscription (DT docs), DPS Hosts capabilities (DT docs), Account Management portal (DT docs). The per-capability schema table is derived from live `dt.system.events` schema inspection on a SaaS tenant (2026-05-19) — the capability-to-unit table is in § 2.

## 2. The DPS Capability Model




DPS bills along **capabilities** — discrete units of platform functionality, each with its own unit of measure and a price defined in the customer's rate card. Capabilities group into categories that match the official documentation taxonomy:

| Category | Capabilities | Unit |
| --- | --- | --- |
| **Host Monitoring** | Full-Stack Monitoring, Infrastructure Monitoring, Foundation & Discovery, Code Monitoring, Kubernetes Platform Monitoring | GiB-hours / host-hours / container-hours / pod-hours |
| **Application Security** | Runtime Vulnerability Analytics (RVA), Runtime Application Protection (RAP) | GiB-hours |
| **Security Posture Management** | Security Posture Management (Kubernetes-scoped) | Record count |
| **Logs powered by Grail** | Ingest & Process, Retain, Retain with Included Queries, Query | Bytes |
| **Events powered by Grail** | Ingest & Process, Retain, Query | Bytes |
| **Metrics powered by Grail** | Ingest & Process (with included-quota subtraction), Retain | Data points / Bytes |
| **Traces powered by Grail** | Ingest & Process, Retain, Query | Bytes ( `ingested_bytes` for Ingest, `billed_bytes` for Retain/Query) |
| **Automation** | Automation Workflow | Workflow-hours (record count per hour) |
| **Standard Function Calls** (formerly AppEngine Functions) | AppEngine Functions - Small (Ad-hoc, Actions, App functions) | Invocations |
| **Synthetic Monitoring** | Browser Monitor / Clickpath, HTTP Monitor | Synthetic actions / HTTP requests |
| **Real User Monitoring** | RUM session billing | Sessions |
| **Files** | Files - Retain | Bytes |
| **Data** | Data Egress (forward logs / metrics / events to AWS S3, GCP, Azure) | Bytes (uncompressed, counted per destination) |

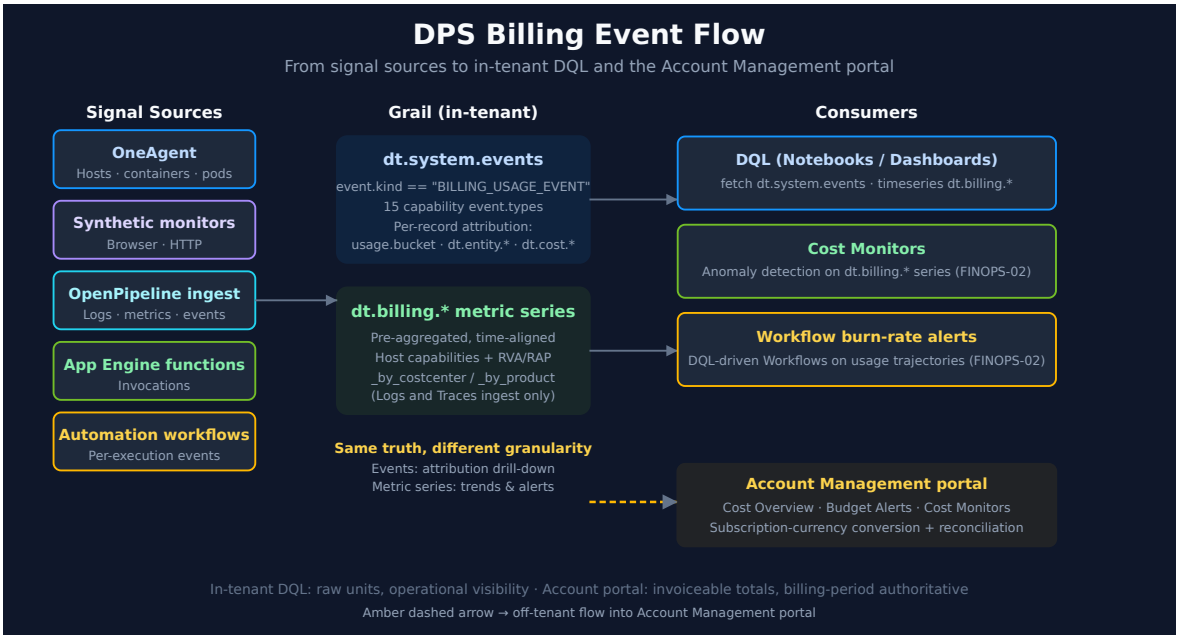
### Why the unit families matter

A common analysis mistake is summing across capabilities — e.g., `total = sum(billed_bytes) + sum(billed_gibibyte_hours)`. The result is dimensionally meaningless. **Always aggregate within a single capability first**, convert to subscription currency at the reporting boundary, and only then combine totals if a single "how much" number is needed.

The conversion factors (units → subscription currency) are in the rate card attached to the customer's DPS agreement and are also surfaced in the Account Management portal's Cost Overview. They are not exposed via DQL — that's a deliberate separation. DQL gives raw consumption; the portal does the currency math.

> **Sources:** Dynatrace Platform Subscription (DT docs), DPS Hosts (DT docs), DPS Application Security (DT docs), DPS Log Management (DT docs), DPS Metrics (DT docs), DPS Traces (DT docs), DPS Automation (DT docs), DPS AppEngine Functions (DT docs). The per-capability category taxonomy matches the layout of the official DPS Usage Details DEMO dashboard.

## 3. Two Data Surfaces — `dt.system.events` and `dt.billing.*`




Consumption appears in two distinct Grail surfaces. They report the same underlying truth but at different granularity. The right choice depends on the question being asked.

## `dt.system.events` — per-record billing events

Every billable activity emits a record into `dt.system.events` with `event.kind == "BILLING_USAGE_EVENT"`. Records carry per-capability attribution (which host, which bucket, which workflow, which synthetic test). Fields vary per `event.type`.

Use this surface when you need:

- Per-host, per-bucket, per-test, per-workflow attribution
- Query-side billing ( `Log Management & Analytics - Query`, `Events - Query`, `Traces - Query`) with `client.application_context` and `user.email`
- Capabilities that don't have a pre-aggregated metric series (Logs - Retain, Events, Metrics - Ingest, Synthetic, RUM, AppEngine, Automation Workflow)
- Custom chargeback expansion via `expand dt.cost.costcenter` / `dt.cost.product`

## `dt.billing.*` — pre-aggregated metric series

For host-based and security capabilities (plus pre-aggregated chargeback views for Logs and Traces), Dynatrace exposes first-class metric series. These are faster to query, naturally time-aligned, and ideal for dashboards and alerts.

| Metric series | Unit | Use |
|---|---|---|
| `dt.billing.full_stack_monitoring.usage` | GiB-hours | Full-Stack hourly trend |
| `dt.billing.infrastructure_monitoring.usage` | host-hours | Infrastructure hourly trend |
| `dt.billing.foundation_and_discovery.usage` | host-hours | Discovery-tier hourly trend |
| `dt.billing.code_monitoring.usage` | container-hours | Code Monitoring hourly trend |
| `dt.billing.kubernetes_monitoring.usage` | pod-hours | K8s Platform hourly trend |
| `dt.billing.runtime_vulnerability_analytics.usage` | GiB-hours | RVA hourly trend |
| `dt.billing.runtime_application_protection.usage` | GiB-hours | RAP hourly trend |

| Metric series | Unit | Use |
| --- | --- | --- |
| `dt.billing.logs.ingest.usage_by_costcenter` | bytes | Log Ingest chargeback by cost center |
| `dt.billing.logs.ingest.usage_by_product` | bytes | Log Ingest chargeback by product |
| `dt.billing.traces.ingest.usage_by_costcenter` | bytes | Trace Ingest chargeback by cost center |
| `dt.billing.traces.ingest.usage_by_product` | bytes | Trace Ingest chargeback by product |

Use this surface when you need:

- Fast hourly / daily trends without raw-record aggregation overhead
- Dashboard tiles and Davis analyzer inputs
- Pre-aggregated cost-center / product views (Logs, Traces)
- Forecast inputs for Davis Predictive AI (FINOPS-02 deep-dives this)

## Decision

In community practice, the split that works is: if your question is *"how much did capability X cost over time?"* — use `dt.billing.*`. If your question is *"who is consuming what?"* or *"which bucket / host / workflow drove the cost?"* — use `dt.system.events`. Many dashboards combine both: `dt.billing.*` for the top-line trend, `dt.system.events` for drill-down attribution.

> **Sources:** Both surfaces verified live on a SaaS tenant (2026-05-19) — the `dt.billing.*` metric catalog returned 13 series; `fetch dt.system.events | filter event.kind == "BILLING_USAGE_EVENT"` returned 15 distinct `event.type` values across 7 unit-field families. § 2 carries the capability-to-unit table.

## 4. Mandatory Patterns — `dedup`, `event.kind`, `billing_type`

Two patterns belong in every DPS consumption query, and a third applies to Standard Function Calls (formerly AppEngine Functions), whose billing events still carry `event.type == "AppEngine Functions - Small"`. Skipping them produces results that look reasonable but are wrong.

### `dedup event.id` before any aggregation

Dynatrace documents the reason: *"Dynatrace refreshes metering records when correcting measurements."* Without dedup, the same consumption period is counted more than once — Dynatrace's own tutorials put the inflation at 10–30%. The canonical pattern is:

```
fetch dt.system.events
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "&lt;capability&gt;"
| dedup event.id      // ← every aggregation chain starts here
| summarize ...
```

Dynatrace's cost tutorials call it mandatory in every billing query. In community practice it is also the pattern most often missing from queries passed around between teams — when adapting a community query, verify the dedup is present.

### Always filter on `event.kind`

`dt.system.events` contains many other event kinds beyond billing. Without `filter event.kind == "BILLING_USAGE_EVENT"`, you'll mix billing records with diagnostic events, audit events, and other platform-internal data.

### `billing_type` on AppEngine — exclude non-billable, keep null

**AppEngine Functions - Small** records can carry a `billing_type`, but the field is optional and is often empty. Exclude records explicitly marked non-billable and keep the ones with no value: `billing_type == "BILLABLE" or isNull(billing_type)`. Dynatrace's own AppEngine consumption queries use exactly that form. A strict `billing_type == "BILLABLE"` drops every record whose field is empty and returns **zero rows with no warning** — on the validation tenant (10/05/2026) all 60 AppEngine records in 30 days had a null `billing_type`, so the strict filter reported no AppEngine usage at all.

### `usage.bucket` vs `usage.event_bucket`

`usage.bucket` is present on every bucket-scoped capability observed, including all three Events capabilities. **Events - Ingest & Process** additionally carries `usage.event_bucket`; Events - Retain and Events - Query do not. Group by `usage.bucket`; when unifying across capabilities, `coalesce(usage.bucket, usage.event_bucket)` is a safe fold.

> **Sources:** Trace a cost spike to its root cause (DT docs) — *"Dynatrace refreshes metering records when correcting measurements."*; every billing query in that tutorial and in Forecast costs with run-rate projections (DT docs) applies `filter event.kind == "BILLING_USAGE_EVENT"` and `dedup event.id`. Standard Function Calls consumption (DT docs) — filter on `event.type == "AppEngine Functions - Small"` *"to find Standard Function Call invocations"*, and both example queries filter on `(billing_type == "BILLABLE" OR isNull(billing_type))`. Field coverage read on a SaaS tenant 10/05/2026 (30 days): AppEngine `billing_type` null on 60/60 records; `usage.bucket` on 3,185/3,185 Events Ingest, 1,984/1,984 Events Retain and 8,871/8,871 Events Query records; `usage.event_bucket` on Events Ingest only.

## 5. Querying Host-Based Capabilities

Host-based capabilities are the easiest to query because of the pre-aggregated `dt.billing.*` metric series. For dashboards and trend views, prefer `timeseries` over `fetch dt.system.events` — it is faster, time-aligned, and the surface FINOPS-02 feeds to Davis forecast and anomaly analyzers.

**Worked example — Full-Stack Monitoring hourly usage:**

```
// Full-Stack Monitoring — hourly GiB-hours over the last 24 hours
timeseries hourlyUsage = sum(dt.billing.full_stack_monitoring.usage, rate:1h),
  from:-24h, interval:1h
```

The `rate:1h` parameter normalizes the metric to hourly rate regardless of the underlying ingestion interval. This is the canonical hourly-trend pattern from the official DEMO dashboard.

**Worked example — all host categories combined for a single hourly comparison:**

```
// All host-based capabilities, side-by-side hourly trend
timeseries {
  full_stack = sum(dt.billing.full_stack_monitoring.usage, rate:1h),
  infrastructure = sum(dt.billing.infrastructure_monitoring.usage, rate:1h),
  discovery = sum(dt.billing.foundation_and_discovery.usage, rate:1h),
  code = sum(dt.billing.code_monitoring.usage, rate:1h),
  k8s = sum(dt.billing.kubernetes_monitoring.usage, rate:1h)
  },
  from:-7d, interval:1h
```

**Worked example — per-host attribution for Full-Stack (which hosts are driving cost):**

> **SaaS 1.347 — staged tenant rollout from 09/16/2026:** billing usage events are moving their host ID from `dt.entity.host` to `dt.smartscape.host`. Verbatim: *"If you use custom DQL queries that reference entity ID attributes in billing usage events, review and update them to the new Smartscape attribute names."* The query below reads `coalesce(toString(dt.smartscape.host), dt.entity.host)`, so it returns the same hosts before and after the change reaches your tenant (on the validation tenant both fields were written, with identical values, on 10/05/2026). What's new in SaaS 1.347 (DT docs). One trap when you filter on the new field: `dt.smartscape.host` is a `smartscape_id`, not a string, so `dt.smartscape.host == "HOST-…"` never matches and raises no error. Compare with `toSmartscapeId("HOST-…")`, or compare `toString(dt.smartscape.host)` (verified 10/05/2026: the string comparison returned `false` and both alternatives `true` for the same host).

```
// Top-spending hosts by Full-Stack Monitoring (uses per-record events for
entity attribution)
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT" and event.type == "Full-Stack
Monitoring"
| dedup event.id
// SaaS 1.347 moves the host ID to dt.smartscape.host; coalesce reads
whichever field your tenant writes.
| fieldsAdd host = coalesce(toString(dt.smartscape.host), dt.entity.host)
| summarize { total_gib_hours = sum(billed_gibibyte_hours) }, by:{ host }
| sort total_gib_hours desc
| limit 10
```

Notice the pattern split — when you need *trend* (line/bar over time), use `dt.billing.*`. When you need *attribution* (which entity drives cost), use `dt.system.events`. The two queries complement each other; many dashboards run both side-by-side.

> **Sources:** DPS Hosts capabilities (DT docs). All three queries verified live on a SaaS tenant (2026-05-19; the trend query re-run 10/05/2026 with its aggregations grouped in `{}`, no notifications); the first query returned real hourly Full-Stack values (e.g., 225.75 GiB-hours/hour during business hours, 162.25 GiB-hours/hour during off-hours).

## 6. Querying Byte-Based Capabilities

Logs, Events, Traces, Files, and Data Egress all bill in bytes. Use `fetch dt.system.events` because the per-bucket (or, for egress, per-destination) attribution is essential for chargeback — but be aware of the field-name variation:

| Capability | Unit field | Bucket field |
|---|---|---|
| Log Management & Analytics - Ingest & Process | `billed_bytes` | `usage.bucket` |
| Log Management & Analytics - Retain | `billed_bytes` | `usage.bucket` |
| Log Management & Analytics - Retain with Included Queries | `billed_bytes` | `usage.bucket` |
| Log Management & Analytics - Query | `billed_bytes` | ( `query_start` instead) |
| Events - Ingest & Process | `billed_bytes` | `usage.bucket` (also `usage.event_bucket`) |
| Events - Retain | `billed_bytes` | `usage.bucket` |
| Events - Query | `billed_bytes` | `usage.bucket` — one record per query per bucket read; time on `query_start` |
| **Traces - Ingest & Process** | **`ingested_bytes`** | `usage.bucket` |

| Capability | Unit field | Bucket field |
|---|---|---|
| Traces - Retain | `billed_bytes` | `usage.bucket` |
| Traces - Query | `billed_bytes` | ( `query_start` instead) |
| Files - Retain | `billed_bytes` | (none — tenant-global) |
| Data Egress | `billed_bytes` | (per-destination; uncompressed) |

**Traces - Ingest is the outlier** — it uses `ingested_bytes` instead of `billed_bytes`. This is the most common cross-capability footgun.

**Worked example — log ingest by bucket over the last 7 days:**

```
// Log Ingest &amp;amp; Process — total GiB ingested per bucket over last 7 days
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "Log Management &amp;amp; Analytics - Ingest &amp;amp; Process"
| dedup event.id
| summarize { total_gib = sum(toDouble(billed_bytes)) / 1073741824 }, by:{
usage.bucket }
| sort total_gib desc
| limit 20
```

The `/ 1073741824` divisor converts bytes → GiB (`1024^3`). On a SaaS validation tenant this returned real per-bucket numbers (e.g., `easytrade` 18.5 GiB / 7 days, `default_logs` 8.8 GiB / 7 days).

**Worked example — Trace Ingest using the `ingested_bytes` field:**

```
// Trace Ingest — note ingested_bytes, NOT billed_bytes
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "Traces - Ingest &amp;amp; Process"
| dedup event.id
| summarize { total_gib = sum(toDouble(ingested_bytes)) / 1073741824 }, by:{
usage.bucket }
| sort total_gib desc
```

**Worked example — hourly log retention bytes per bucket (the canonical area-chart pattern from the DEMO dashboard):**

```
// Log Retain — hourly bytes per bucket, area-chart input
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "Log Management &amp;amp; Analytics - Retain"
| dedup event.id
| fieldsAdd billing_period = bin(timestamp, 1h)
| fields billing_period, billed_bytes, usage.bucket = coalesce(usage.bucket,
usage.event_bucket)
| makeTimeseries billed_bytes = max(billed_bytes), by:{ usage.bucket }, time:
billing_period, interval:1h
```

Retention queries use `max(billed_bytes)` rather than `sum()` because each hour's record carries the *cumulative* bucket size — taking the max within the bucket gives the high-water mark for that hour. Summing would double-count.

## Data Egress — forwarding data out of Dynatrace

**Data Egress** is a distinct DPS capability that bills when you forward data (logs, metrics, or events) *out* of Dynatrace to an external destination — currently **AWS S3, GCP, and Azure**. It bills in bytes ( `billed_bytes` ) measured on the **uncompressed** size, and is counted **once per destination**: routing the same data to N destinations multiplies consumption N×. List price is **$0.15 / GiB** at the time of writing — your rate card may differ.

```
// Data Egress — total GiB forwarded to external destinations over last 7 days
// Doc-cited shape (event.type "Data Egress", billed_bytes); validate in your
tenant once egress is configured
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "Data Egress"
| dedup event.id
| summarize { total_gib = sum(toDouble(billed_bytes)) / 1073741824 }
```

The account-level equivalent is **Account Management → Subscription → Overview → Cost and usage details → Usage summary → Data Egress**, or the Account Management API `GET /subscriptions/{subscriptionId}/usage`. Because forwarding *is* the cost driver, the optimization levers are *filter before you forward* and *minimize the destination count* — the same shape as the FINOPS-03 Cut / Tune / Filter framework.

> **Sources:** DPS Log Management (DT docs), DPS Events (DT docs), DPS Traces (DT docs), Data Egress (DT docs). The hourly-area-chart pattern is lifted verbatim from the DPS Usage Details DEMO

> dashboard (Log-Retain tile). The Log / Trace / Retain queries are verified live on a SaaS tenant (2026-05-19); the **Data Egress query is doc-cited, not live-validated** — egress was not configured on the validation tenant, so verify the `event.type` and `billed_bytes` shape in your own tenant once forwarding is enabled.

## 7. Querying Count-Based Capabilities

Synthetic, RUM, AppEngine, and Automation Workflow bill on counts rather than bytes or hours. The query shape is similar across all four — replace the unit field appropriately.

| Capability | Unit field | Attribution |
|---|---|---|
| Browser Monitor / Clickpath | `billed_synthetic_action_count` | `dt.entity.synthetic_test` |
| HTTP Monitor | `billed_http_request_count` | `dt.entity.http_check` |
| Real User Monitoring | `billed_sessions` | `dt.entity.application`, `device.type` |
| AppEngine Functions - Small | `billed_invocations` | `user.email`, `caller.app.id` |
| Automation Workflow | (record count — 1 per standard workflow per hour of existence) | `workflow.title`, `workflow.owner`, `workflow.trigger_type` |

**Automation Workflow records are workflow-hours, not executions.** A deployed standard workflow is billed for every hour it exists, whether it runs once a month or every minute, so execution frequency does not change this capability. Executions are billed elsewhere: each task run is a Standard Function Call invocation, and any DQL a task runs bills the query capability it reads. Simple and draft-only workflows consume no workflow-hours.

**Worked example — workflow-hours by trigger type:**

```
// Automation Workflow — workflow-hours by trigger type (one record = one
standard workflow existing for one hour)
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT" and event.type == "Automation
Workflow"
| dedup event.id
| summarize { workflow_hours = count(), workflows =
countDistinctExact(workflow.id) }, by:{ workflow.trigger_type }
| sort workflow_hours desc
```

**Worked example — workflow-hours by owner (who owns the deployed standard workflows):**

```
// Workflow-hours by owner — candidates for review: delete unused standard
workflows, or convert to simple / draft
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT" and event.type == "Automation
Workflow"
| dedup event.id
| summarize { workflow_hours = count(), workflows =
countDistinctExact(workflow.id) }, by:{ workflow.owner, workflow.trigger_type
}
| sort workflow_hours desc
| limit 10
```

**Worked example — Standard Function Calls by function type excluding records marked non-billable:**

```
// AppEngine Functions - Small — invocation count by function type
// (Ad-hoc = workflow-task invocations, Actions = workflow actions, App
functions = standalone app code)
fetch dt.system.events, from:-7d
| filter event.type == "AppEngine Functions - Small"
| filter event.kind == "BILLING_USAGE_EVENT"
| filter billing_type == "BILLABLE" or isNull(billing_type)   // the field is
often empty — keep nulls
| dedup event.id
| fieldsAdd category = if(matchesPhrase(function.type, "AD_HOC"), "Ad-hoc",
                      else: if(matchesPhrase(function.type, "ACTION"),
"Actions",
                      else: if(matchesPhrase(function.type, "STANDARD"), "App
functions")))
| summarize { invocations = sum(coalesce(toLong(billed_invocations), 1)) },
by:{ category }
| sort invocations desc
```

**Worked example — synthetic action consumption by test:**

```
// Synthetic Browser / Clickpath — action count by test (high-frequency
monitors are review candidates)
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT" and event.type == "Browser
Monitor or Clickpath"
| dedup event.id
| summarize { actions = sum(toLong(billed_synthetic_action_count)) }, by:{
dt.entity.synthetic_test }
| sort actions desc
| limit 10
```

> **Sources:** DPS Automation (DT docs), DPS AppEngine Functions (DT docs). Calculate your consumption of Automation Workflow (DT docs) — *"Workflow hours are the number of hours that a workflow has existed in your environment, measured since the point of its creation."*; *"Simple workflows do not directly consume workflow hours"*; *"Each workflow execution triggers Standard Function Calls that execute tasks"* Standard Function Calls consumption (DT docs) — the `billing_type` filter, `function.type` category mapping and `user.email` / `caller.app.id` attribution. Executed on a SaaS tenant 10/05/2026: every Automation Workflow record is unique per `workflow.id` and hour (21,954 workflow-hours, maximum 1 per hour); manual 10,920 workflow-hours / 65 workflows, schedule 8,232 / 49, event 2,802 / 17 over 7 days — 168 hours per workflow, run or not; the AppEngine query returned 60 Ad-hoc invocations over 30 days, where the strict `billing_type == "BILLABLE"` form returned none. Synthetic query verified 2026-05-19.

## 8. Metrics-Ingest with Included-Quota Subtraction

Metrics ingestion is the most subtle capability because it has **included quotas**. Each GiB of memory monitored under Full-Stack Monitoring grants 900 included metric data points per 15-minute interval; each host monitored under Infrastructure Monitoring grants 1,500. Billable Metrics-Ingest is *only the portion of total data points that exceeds the included allowance*.

This is the canonical query from the official DEMO dashboard — annotated:

```
// Metrics-Ingest — billable usage after included-quota subtraction
// (canonical pattern; tenant must have ≥4 hours of data; metrics billing lags
by ~4 hours)
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT" and event.type == "Metrics -
Ingest &amp;amp; Process"
| dedup event.id
| fieldsAdd monitoring_source = if(monitoring_source == "fullstack" or
monitoring_source == "infrastructure",
                                  monitoring_source, else: "other")
| summarize { total_data_points = toLong(sum(data_points)) }, by:{
usage.start, monitoring_source }
| makeTimeseries { total_usage = sum(total_data_points, default: 0) },
    interval:15m, time: usage.start, by:{ monitoring_source }
// Join in the included-quota allowance from dt.billing.*
// dt.billing.*.usage summed per 15-min bucket = GiB-hours (host-hours) in
that bucket; × 4 → GiB (hosts);
// × 900 (Full-Stack, per GiB) / × 1500 (Infrastructure, per host) = included
points per 15 min
| join [
    timeseries { included_usage = sum(dt.billing.full_stack_monitoring.usage,
default: 0) },
      interval:15m, nonempty:true
    | fields monitoring_source = "fullstack", included_usage = 4 * 900 *
included_usage[]
    | append [
        timeseries { included_usage =
sum(dt.billing.infrastructure_monitoring.usage, default: 0) },
          interval:15m, nonempty:true
        | fields monitoring_source = "infrastructure", included_usage = 4 *
1500 * included_usage[]
      ]
  ], on:{ monitoring_source }, fields:{ included_usage }, kind:leftOuter
| fieldsAdd billed_usage = if(isNotNull(included_usage) and total_usage[] &gt;
included_usage[],
                            total_usage[] - included_usage[], else:0)
| fieldsAdd billed_usage = if(isNull(included_usage), total_usage, else:
billed_usage)
| fieldsAdd billed_usage = arraySum(billed_usage)
| fieldsKeep monitoring_source, billed_usage
```

**Key elements:**

1. `monitoring_source` **rollup** — collapses fine-grained values into `fullstack`, `infrastructure`, or `other`. Other-source data points (e.g., custom metrics not tied to a host) have no included quota and are billed directly.

2. **15-minute interval** — matches Dynatrace's billing-event window. Don't change this.
3. **The `4 * 900` and `4 * 1500` factors** — a 15-minute bucket of `dt.billing.*.usage` holds GiB-hours (Full-Stack) or host-hours (Infrastructure); `× 4` converts that back to GiB or hosts. `900` is the included data points per GiB of memory per 15-minute interval for Full-Stack; `1500` is the included data points per host per 15-minute interval for Infrastructure. These constants are part of the platform billing model and should not be tuned per-customer.
4. **`kind: leftOuter`** — `other`-source data has no matching included quota; `leftOuter` keeps those rows with `null` included_usage; the next two `fieldsAdd` lines handle that case.
5. **The 4-hour lag** — Metrics-Ingest's `usage.start` is approximately 4 hours behind `timestamp`. Query timeframes shorter than 4 hours will show incomplete data. Always query at least the last 4 hours.

> **Sources:** Pattern lifted from the DPS Usage Details DEMO dashboard Metrics tile — verbatim including the `4 * 900` / `4 * 1500` constants. DPS Metrics (DT docs). Full-Stack Monitoring (DT docs) — *"Every contributing GiB of host or application memory adds 900 custom metric data points in each 15-minute interval."* Infrastructure Monitoring (DT docs) — *"For each host, 1,500 custom metric data points are included in each 15-minute interval."* Re-executed 10/05/2026 (7 days, no notifications): `fullstack` billed 0, `other` billed 43,195,590 data points. The 4-hour lag caveat is verbatim from the demo dashboard's documentation tile. **Softened:** the `900` / `1500` constants are platform-billing-model values that may change in future DPS rate-card revisions — verify against current docs before relying on them in production reporting.

## 9. Query-Side Billing — Chargeback by App and User

Three Grail capabilities bill on **query execution** in addition to ingest and retention: Log Management, Events, and Traces. The query-side records carry rich attribution — which app made the query, which user, even which finer-grained source.

They are not the only query-side event type. The billing usage event model also defines `Digital Experience Monitoring - Query` — *"a billing usage event of a Digital Experience Monitoring query execution"* — with the same `query_id` / `billed_bytes` / `usage.bucket` shape, although the DPS capability list does not name it as a rate-card

line. On a validation tenant (10/14/2026) it carried 29 records in 30 days, all reads of the synthetic-event and synthetic-user-session buckets. When you audit query-side consumption, filter on `contains(event.type, "Query")` rather than on the three names, so this type is not missed.

> **Sources:** System event models (DT docs) — *"Model describing a billing usage event of a Digital Experience Monitoring query execution."* Capabilities (DT docs) — lists Log Management & Analytics, Traces, Events and Metrics - Query, and no Digital Experience Monitoring - Query line.

| Field | Meaning |
|---|---|
| `query_start` | When the query ran (use this for time-bucketing, not `timestamp`) |
| `client.source` | What issued the query — a dashboard URL, an anomaly detector, an automation workflow, or a platform service |
| `client.application_context` | Which app made the query — may be empty on billing records (see below) |
| `query_id` | The query; an Events - Query record is one query × one bucket, so count distinct `query_id`, not records |
| `user.email` | Who ran the query |
| `billed_bytes` | Bytes scanned by the query |

This is the foundation for chargeback by team — a Notebook that scans 50 GiB hourly is a different financial conversation than the same DQL pasted into a one-time investigation.

**Worked example — top query-side consumers by source and user (Logs and Events):**

```
// Top query-side consumers — bytes scanned by source and user (Logs and
Events)
fetch dt.system.events, from:-30d
| filter event.kind == "BILLING_USAGE_EVENT"
| filter in(event.type, {"Log Management &amp;amp; Analytics - Query", "Events -
Query"})
| dedup event.id
| summarize {
    gib_scanned = sum(toDouble(billed_bytes)) / 1073741824,
    query_count = countDistinctExact(query_id)
  }, by:{ event.type, client.source, user.email }
| sort gib_scanned desc
| limit 20
```

**Worked example — query-cost trend over time (use `query_start`, not `timestamp`):**

```
// Hourly log-query bytes scanned — bucketed by query_start (NOT timestamp)
fetch dt.system.events, from:-7d
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "Log Management &amp;amp; Analytics - Query"
| dedup event.id
| summarize { data_read_bytes = sum(billed_bytes) }, by:{ startHour =
bin(query_start, 1h) }
| sort startHour asc
```

The query above already covers Events; add Traces with `event.type == "Traces - Query"`. Two schema differences matter when you read the result. An **Events - Query** record is one query × one bucket — on the validation tenant one service user had 408 records for 204 queries over 30 days — so count `countDistinctExact(query_id)`, not records. **Log Management & Analytics - Query** records carry no bucket at all.

**`client.source` vs `client.application_context`.** `client.application_context` is documented on the billing event model but may be empty: on the validation tenant (10/05/2026) it was null on every one of 19,156 Log, Events and Digital Experience Monitoring query billing records in 30 days, and populated only on `QUERY_EXECUTION_EVENT` records (in reverse-DNS form such as `dynatrace.dashboards`). `client.source` carried the attribution on 10,277 of 10,280 Log Query and 8,460 of 8,871 Events Query records — there, one static-threshold anomaly detector accounted for 534 GiB of the 535 GiB of log-query scan. Dynatrace's own spike tutorial groups query billing by `client.source` for the same reason. A **null** `client.source` is not an error: it marks a query with no source context, typically platform-service traffic.

This is also how you settle the question *"does the Distributed Tracing or Services app consume Query DPS?"* for your own estate — swap the filter to `Traces - Query` and group by `client.source`. Two cautions before you read the result as an answer: query-side records are written **only when the capability is actually read**, so an empty result means nobody queried traces in the window, not that the app is free; and a tenant with little trace-query traffic can return nothing at all over 30 days. Widen the timeframe and confirm the capability is in use before drawing a conclusion either way.

> **Sources:** Trace a cost spike to its root cause (DT docs) — *"For query billing events, client.source identifies who is scanning data."* Field coverage and the query above executed on a SaaS tenant 10/05/2026 (30 days; top row: Log Query, static-threshold detector, 534.3 GiB over 10,258 queries). DPS Log Management (DT docs), DPS Events (DT docs), DPS Traces (DT docs). Query-side billing fires only when the capability is in active use — on validation tenants with light read traffic, these queries may return zero rows. Verify in your tenant before relying on them for chargeback reporting.

## 10. Per-Bucket and Per-Cost-Center Attribution

> **Davis AI events in custom buckets are now usage-tracked (SaaS 1.345, 08/11/2026 — staged rollout).** Reported as: Davis AI events are usage-tracked when routed to a bucket other than `default_davis_events`. **Flagged 09/02/2026 — could not be re-verified.** The wording was recorded here as a verbatim quote from the SaaS 1.345 release note, but it is not on that page today (checked two ways: text extraction and a rendered fetch), nor on 1.344, 1.346, or the DPS log-management page. The behaviour is plausible and the operational guidance below stands on its own, but **treat the sourcing as open** — confirm against your own consumption data before relying on the version gate. If you route Davis events to a custom bucket — a common move for retention or access-scoping — that traffic previously escaped ingest usage tracking and now appears in it. **Expect reported usage to rise without ingest volume changing**; the events were always being ingested, they just were not being counted. Budget alerts and month-over-month comparisons calibrated before the rollout will read as a step change, so re-baseline rather than investigating a phantom spike. A transient accounting artifact was also recorded here — one 15-minute timeframe showing less usage and the next showing more, with the total unchanged. **Same sourcing caveat as above:** this was recorded as a verbatim

> quote from the same release note and could not be re-verified there on 09/02/2026. Davis events left in `default_davis_events` are unaffected.
>
> Source: What's new in Dynatrace SaaS 1.345 (DT docs)

Three attribution surfaces exist, in increasing order of attribution power:

1. **`usage.bucket` / `usage.event_bucket`** — every byte-based capability includes the bucket name. Effective only if your bucket-naming convention encodes team / product / environment (covered in ORGNZ).
2. **`dt.cost.costcenter[]` / `dt.cost.product[]` arrays** — present on Metrics, Logs - Retain, and a few other capabilities. Each record carries an array of `{key, billed_bytes}` (or `{key, data_points}`) tuples — one per cost center or product the usage is allocated to.
3. **Pre-aggregated `dt.billing.*_by_costcenter` / `_by_product` metric series** — Dynatrace pre-computes the cost-center / product attribution for **Logs - Ingest** and **Traces - Ingest** and exposes them as standard metric series. Fastest to query.

**Worked example — hourly log ingest by cost center using the pre-aggregated metric:**

```
// Pre-aggregated log ingest by cost center — fast chargeback view
timeseries logs_by_cc = sum(dt.billing.logs.ingest.usage_by_costcenter),
  from:-7d, interval:1h, by:{ dt.cost.costcenter }
```

**Worked example — manual cost-center attribution via `expand`:**

When the pre-aggregated metric isn't available (Events, Logs - Retain, etc.), expand the per-record array manually:

```
// Manual cost-center attribution — Log Retain via expand
// Retain records are hourly snapshots of a cumulative size: sum across
buckets within an hour, then average the hours
fetch dt.system.events, from:-24h
| filter event.kind == "BILLING_USAGE_EVENT"
| filter event.type == "Log Management &amp;amp; Analytics - Retain"
| dedup event.id
| expand dt.cost.costcenter
| fieldsAdd costcenter = dt.cost.costcenter[`key`],
            cc_bytes = toDouble(dt.cost.costcenter[`billed_bytes`]),
            hour = bin(timestamp, 1h)
| summarize { hourly_bytes = sum(cc_bytes) }, by:{ costcenter, hour }
| summarize { avg_retained_gib = avg(hourly_bytes) / 1073741824 }, by:{
costcenter }
| sort avg_retained_gib desc
```

Retain is the case where pitfall #6 bites after an `expand`: each record is an hourly snapshot of the bucket's retained size, so summing a day of records multiplies the stored volume by 24. On a validation tenant (10/05/2026) this query returned two cost centers: `unassigned` (~16 TiB retained on average over the day) and `not-allowlisted` (~0.4 TiB); a plain `sum()` over the same day read 383,119 GiB — 24× too high. The `unassigned` bucket is universal — every tenant has it, and reducing its share by populating proper cost-center labels at ingest time is the upstream lever.

**Defining cost-center / product labels:** The values come from data ingested with `dt.cost.costcenter` and `dt.cost.product` attributes — typically applied via OneAgent host properties (covered in the FAQ series entry on tagging sources, standards, and strategy) or via OpenPipeline enrichment rules at ingest. Where host-group granularity is enough, Dynatrace names a preferred route: *"If cost attribution at the host group level is sufficient, the preferred approach is to map deployment-level primary Grail fields such as dt.host_group.id or host.name to dt.cost.costcenter and dt.cost.product via the Cost Allocation stage in OpenPipeline."* Use host tags or `DT_TAGS` when you need finer granularity than the host group. Reducing the `unassigned` share is part of the FINOPS-03 optimization framework.

> **Sources:** Configure cost allocation (DT docs) — the host-group route quoted above. Allocate your DPS costs (DT docs) — tags *"flow through Grail as dt.cost.costcenter and dt.cost.product attributes"*. No docs page found documents the `dt.billing.logs.ingest.usage_by_costcenter` and `dt.billing.traces.ingest.usage_by_costcenter` metric series; both keys were observed in the tenant metric catalog ( `metrics | filter startsWith(metric.key,`

> `"dt.billing")` ), read 10/14/2026. Both queries verified live on a SaaS tenant (2026-05-19); the manual-expand query re-executed 10/05/2026 (no notifications) with the results above.

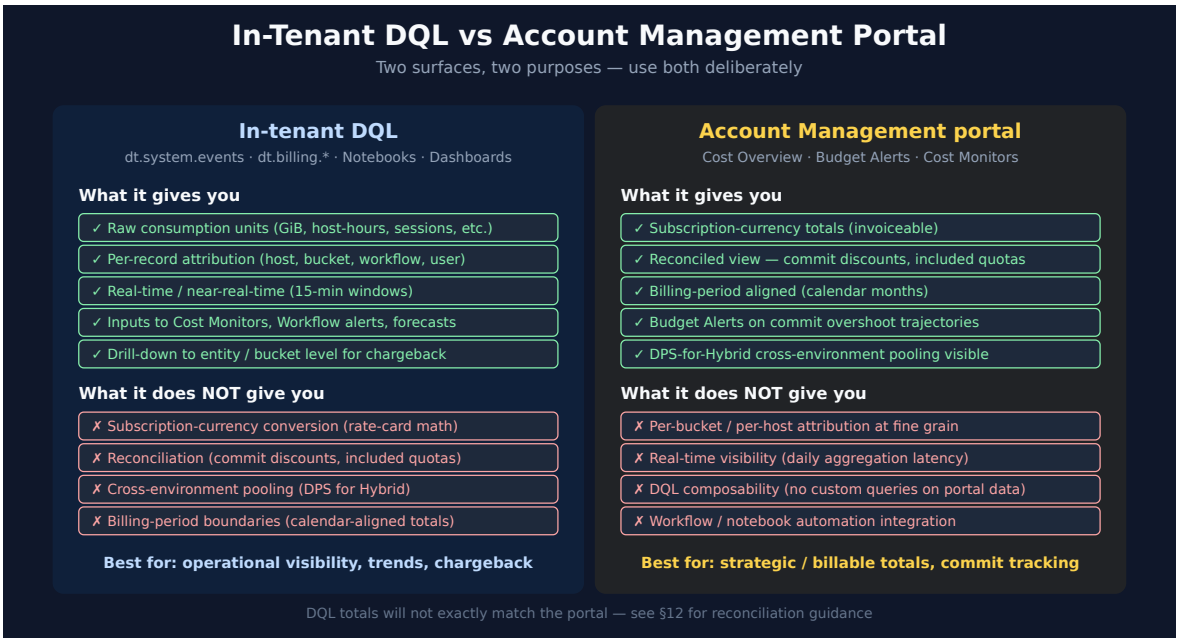
## 11. Common Pitfalls

| # | Pitfall | What goes wrong | Fix |
|---|---|---|---|
| 1 | Missing `dedup event.id` | The same consumption period counted more than once when metering records are refreshed — 10–30% high | Add `dedup event.id` immediately after `filter event.kind == "BILLING_USAGE_EVENT"` |
| 2 | Assuming `billed_bytes` is universal | Trace Ingest returns NULL — query silently shows zero | Trace Ingest uses `ingested_bytes` ; check the unit for each capability in § 2 |
| 3 | Summing across capabilities | Combining `billed_bytes` + `billed_gibibyte_hours` produces dimensional nonsense | Aggregate within one `event.type` , convert to currency at the reporting boundary |
| 4 | Filtering on `dt.security_context` for attribution | The field is literally `"BILLING_USAGE_EVENT"` on every record | Use `usage.bucket` , `dt.entity.host` (moving to `dt.smartscape.host` with SaaS 1.347), or `dt.cost.costcenter[]` for attribution |
| 5 | Short timeframes on Metrics-Ingest | Last 4 hours are incomplete — under-reports consumption | Always query Metrics-Ingest with at least 4 hours of timeframe |
| 6 | Using `sum()` on retention bytes | Each hourly record carries the cumulative bucket size; summing inflates by 24× per day | Use `max(billed_bytes)` for Retain event types; `sum()` is correct for Ingest |

| # | Pitfall | What goes wrong | Fix |
|---|---|---|---|
| 7 | Forgetting the included-quota subtraction for Metrics-Ingest | Over-reports billable metric consumption by the included allowance | Use the canonical query in §8; the included-quota math is non-trivial |
| 8 | Bucketing query-side billing on `timestamp` | Misaligns with actual query execution time | Bucket on `query_start` for query-side billing events |
| 9 | Grouping Events Retain / Query by `usage.event_bucket` | A single null row — only Events Ingest carries `usage.event_bucket` | Group by `usage.bucket`; `coalesce(usage.bucket, usage.event_bucket)` for cross-capability work |
| 10 | Strict `billing_type == "BILLABLE"` on AppEngine | The field is often null, so the filter drops every record and reports zero usage | `billing_type == "BILLABLE" or isNull(billing_type)` — Dynatrace's own AppEngine queries keep nulls |

> **Sources:** Trace a cost spike to its root cause (DT docs) — lists *"Missing dedup event.id"* as a common investigation mistake (pitfall #1). Pitfalls #2–#8 come from live tenant validation against the DPS Usage Details DEMO dashboard's queries (2026-05-19); #9 and #10 were re-read on a SaaS tenant 10/05/2026, and #10's fix is the filter in Standard Function Calls consumption (DT docs).

# 12. Validating Numbers Against Account Management




DQL totals will not exactly match the Subscription portal's Cost Overview. This is not a bug — they measure overlapping but distinct things. Three reconciliation factors:

## Factor 1 — Subscription-currency conversion

DQL returns **raw units** (GiB, host-hours, sessions, invocations). The portal applies **rate-card conversion** to express usage in subscription currency. The conversion factors are not exposed in DQL by design — they belong to the agreement, not the tenant.

Host capabilities are metered in 15-minute intervals, and an interval with less than 15 minutes of monitoring is rounded up to 15 minutes. The rounding is per 15-minute interval — the same interval the in-tenant host records cover — so it is reflected in `billed_gibibyte_hours` rather than applied later in the portal, and it is not a source of DQL-vs-portal drift.

**Practical implication:** If you need a subscription-currency total, the portal's Cost Overview is the authoritative source. DQL is for *operational visibility* (what's consuming what, where is the trend going), not *invoiceable totals*.

## Factor 2 — Reconciliation lag

The portal applies post-hoc reconciliation: included quotas, cross-environment commits (DPS for Hybrid), and certain capability-specific true-ups. These adjustments land in the portal but do not modify the per-record `dt.system.events` data.

**Practical implication:** Daily DQL totals will usually be *higher* than the portal's recognized billable totals — DQL counts everything; the portal subtracts included quotas and may apply commitment discounts.

## Factor 3 — Time-window alignment

The portal aligns to billing-period boundaries (typically calendar months in the customer's contract time zone). DQL aligns to whatever time range you query. For monthly reconciliation, query `from:` and `to:` exactly matched to the billing-period start/end in the contract time zone.

Additionally, **Metrics-Ingest has a ~4-hour data lag** — the portal incorporates a corrected view once the lag closes, but a same-day DQL total will under-report metrics specifically.

## When the gap is large

If your DQL total deviates from the portal by more than ~15% for the same period after accounting for the three factors above, investigate:

- Are you missing capabilities? List the `event.type` values your tenant actually emits — `fetch dt.system.events, from:-30d | filter event.kind == "BILLING_USAGE_EVENT" | summarize n = count(), by:{event.type}` — and check that each one is covered by a query.
- Is `dedup event.id` present? Missing dedup inflates totals — Dynatrace's tutorials put it at 10–30%.
- Are retention queries using `sum()` where they should use `max()`? (Pitfall #6.)
- For Metrics-Ingest, did you subtract included quotas? (Section 8.)
- For Traces - Ingest, are you reading `ingested_bytes`? Reading `billed_bytes` returns nothing.

> **Sources:** Account Management portal (DT docs), Forecast costs with run-rate projections (DT docs) — the 10–30% dedup figure. Full-Stack Monitoring (DT docs) — *"When a host or container is*

> *monitored for fewer than 15 minutes in an interval, GiB-hour consumption is rounded up to 15 minutes before consumption is calculated."* **Derived:** Full-Stack billing records span 900 s, and on a SaaS tenant (10/05/2026) their 24-hour sum (4,026 GiB-h) matched `dt.billing.full_stack_monitoring.usage` (3,984 GiB-h) within about 1%, so the rounding is in the records DQL reads. **Softened:** the specific reconciliation factors the portal applies (commitment discounts, true-ups, DPS-for-Hybrid pooling) evolve per-contract — the three factors above are the generally observable categories, not an exhaustive list of every portal-side adjustment.

## 13. Bootstrap from the Ready-Made Usage Dashboards

Dynatrace ships ready-made **usage dashboards** built on the same billing events this notebook queries: **Usage - Overview** (every rate-card capability, trend, and cost-center breakdown) plus drill-downs — **Usage - Traces**, **Usage - Logs**, **Usage - Metrics**, and **Usage - Full-Stack**. They are the fastest way to get started:

1. In Dynatrace, go to **Dashboards** → **Ready-made** and search for **Usage - Overview**. Viewing them needs `storage:system:read`.
2. Ready-made dashboards are read-only — save a copy to get an editable version in your tenant.
3. Customize the cost-center / product groupings to match your bucket-naming or `dt.cost.*` labels.
4. Set the timeframe to align with your billing period (see §12).
5. Use the dashboards as the operational view; use this notebook as the query reference when you need to extend or debug.

The queries in §§5–9 were originally adapted from the tiles of the **DPS Usage Details DEMO dashboard**, which earlier revisions of this entry described as linked from the License overview. That page (re-read 09/28/2026) no longer links it, so treat the ready-made Usage dashboards above as the maintained starting point and cross-check the §§5–9 patterns against their tiles when they change.

> **Sources:** Ready-made usage dashboards (DT docs) — the data *"includes billing usage events and query execution events"*; Ready-made dashboards (DT docs) — *"Save a copy and customize your*

> *copy".*

## 14. Recommended Approach

A workable plan for putting in-tenant consumption visibility in place:

1. **Start with the ready-made Usage dashboards** (§13). Save a copy, customize it, and make it your team's standing consumption view — the one you open most days. Set its refresh cadence to match how often the numbers are actually acted on (daily review does not need a 1-minute refresh); FINOPS-03 §4 covers why dashboard refresh rate is itself a consumption lever on Logs, Events, and Traces.
2. **Adopt the three mandatory patterns** (§4) — `dedup event.id`, `event.kind` filter, and on AppEngine `billing_type == "BILLABLE" or isNull(billing_type)`. Audit any consumption query in your tenant against these before trusting its numbers.
3. **Use the right surface for the question**: `dt.billing.*` for trends and Davis analyzer inputs; `dt.system.events` for attribution and chargeback.
4. **Pre-aggregated chargeback first.** For Logs and Traces ingest, use `dt.billing.*_by_costcenter` / `_by_product` before reaching for `expand dt.cost.costcenter`. It's faster and cleaner.
5. **Set realistic reconciliation expectations.** DQL totals are not invoiceable. The portal is authoritative for billing-period totals; DQL is authoritative for operational visibility.
6. **Reduce the `unassigned` cost-center share** at ingest time, not at query time. The OpenPipeline / OneAgent enrichment patterns covered in ORGNZ and FAQ-02 are the upstream lever.
7. **Pair with FINOPS-02 for forecasting** (when will we hit commit?) and **FINOPS-03 for optimization** (what should we cut, tune, or filter?). FINOPS-01 is the *what is being used* foundation; the next two entries are *what will be used* and *what should change*.

## Summary

DPS consumption lives in two places — per-record in `dt.system.events` and pre-aggregated in `dt.billing.*`. The schema is per-capability with seven distinct unit-field families, and the three non-negotiable patterns (`dedup event.id`, `event.kind` filter, and the null-tolerant `billing_type` filter on AppEngine) appear in every well-formed query. The ready-made Usage dashboards are the bootstrap; the queries in §§5–9 are the

building blocks. DQL totals will not exactly match the Subscription portal, and that is by design — the portal does subscription-currency conversion and reconciliation that DQL does not. Use both surfaces deliberately.

## Next Steps

- Read **FINOPS-02** for forecasting and anomaly detection on top of the queries in this entry — including Cost Monitors, Davis Predictive AI on `dt.billing.*` metrics, and Workflow-based burn-rate alerting.
- Read **FINOPS-03** for the Cut / Tune / Filter optimization decision framework — once you know what's being consumed (FINOPS-01), the next question is what to do about it.
- Read the **ORGNZ** topic series for bucket strategy and retention policy design — the upstream lever for byte-based capability attribution.
- Read **OPLOGS** for OpenPipeline sampling and filter patterns — the upstream lever for log ingest reduction.
- Open the Account Management portal and run the Cost Overview alongside your DQL totals for one billing period to develop intuition for the reconciliation factors in §12.

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*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*